



Indian Institute of Information Technology, Lucknow

An Institute of National Importance by Act of Parliament
Chak Ganjaria (C.G.) City, Lucknow – 226002, (U.P.) – India

भारतीय सूचना प्रौद्योगिकी संस्थान लखनऊ

संसद के अधिनियम द्वारा अधिकृत राष्ट्रीय महत्व का संस्थान

चक गंजरिया (सी.जी.) सिटी, लखनऊ २२६००२ - (उत्तर प्रदेश), भारत

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Department of Computer Science

Semester: III, VII

Course Code:

Course Name: Optimization Techniques

Credits	L	T	P	Section (Group)
4	3	0	1	B.Tech, M.Tech, M.Sc

Course Module Details

Objective(s):

To introduce the mathematical foundations and computational techniques of linear and nonlinear optimization, and to develop the ability to formulate, analyze, and solve constrained and unconstrained optimization problems using classical and modern algorithms.

Pre-Requisites:

Basic concepts of multivariable calculus and matrix algebra are required.

Description:

Unit I: Types of Optimization Problems, Linear Programming, Simplex Method, Artificial Variables, Matrix Form, Revised Simplex Method.

Unit II: Duality, The Duality Theorem, Sensitivity, Dual Simplex Method, Matrix Calculus, Conditions for Solution of an Unconstrained Problem.

Unit III: Fibonacci and Golden Section Search Method, Conjugate Directions, Conjugate Direction Method, Conjugate Gradient Method, Modified Newton Method, Construction of Inverse, Davidon-Fletcher-Powell Method.

Unit IV: Constrained Minimization Conditions, Gradient Projection Method, Penalty Methods, Barrier Methods, Sub-gradient Method.

Unit V: Primal-Dual Methods, Genetic Algorithm.

Laboratory Experiments:

1. Implementation of Simplex Method.
2. Implementation of Revised Simplex Method.
3. Implementation of Duality Theorem Problem.
4. Implementation of Dual Simplex Method.
5. Implementation of an Unconstrained Problem.
6. Implementation of Fibonacci & Golden Section Search.
7. Implementation of Newton's Method.
8. Implementation of Conjugate Direction Method.



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9. Implementation of Conjugate Gradient Method.
10. Implementation of Modified Newton Method.
11. Implementation of Davidon-Fletcher-Powell Method.
12. Implementation of Gradient Projection Method.
13. Implementation of Penalty Method.
14. Implementation of Barrier Method.
15. Implementation of Sub-gradient Method.

Learning Outcomes Expected:

After completing the course, the student will be able to:

- Formulate real-world problems as linear programs and solve them using the Simplex and Revised Simplex methods.
- Apply duality theory and sensitivity analysis to interpret and solve optimization problems.
- Use search and gradient-based methods (Fibonacci, Golden Section, Conjugate Gradient, Newton, DFP) to solve unconstrained optimization problems.
- Handle constrained problems using penalty, barrier, and gradient projection techniques.
- Analyze and implement primal-dual and evolutionary (genetic algorithm) approaches for complex optimization tasks.

Contact Details: Dr. Ramprasad Sarkar, Assistant Professor, IIT Lucknow

Courseware and Reference Books

Textbooks:

- T1. D. G. Luenberger and Y. Ye, Linear and Nonlinear Programming, 3rd Edition, Springer, 2008.
T2. K. Deb, Optimization for Engineering Design: Algorithms and Examples, 2nd Edition, PHI Learning, 2012.

References:

- R1. J. Nocedal and S. J. Wright, Numerical Optimization, 2nd Edition, Springer, 2006.
R2. S. Boyd and L. Vandenberghe, Convex Optimization, 1st Edition, Cambridge University Press, 2015.